

56. Temperature Sensor (TSN)

56.1 Overview

The on-chip Temperature Sensor (TSN) determines and monitors the die temperature for reliable operation of the device. The sensor outputs a voltage directly proportional to the die temperature, and the relationship between the die temperature and the output voltage is fairly linear. The output voltage is provided to the ADC16H for conversion and can be further used by the end application.

Table 56.1 lists the TSN specifications, and Figure 56.1 shows a block diagram.

Table 56.1 TSN specifications

Item	Description
Temperature sensor voltage output	Temperature sensor outputs a voltage to the A/D converter
Module-stop function	Module-stop state can be set to reduce power consumption
Temperature sensor calibration data	Reference data measured for each chip at factory shipment is stored in a register
Temperature monitor reset source	The sensor outputs an abnormal temperature detection signal to the reset control circuit
TrustZone Filter	Security and Privilege attribution can be set

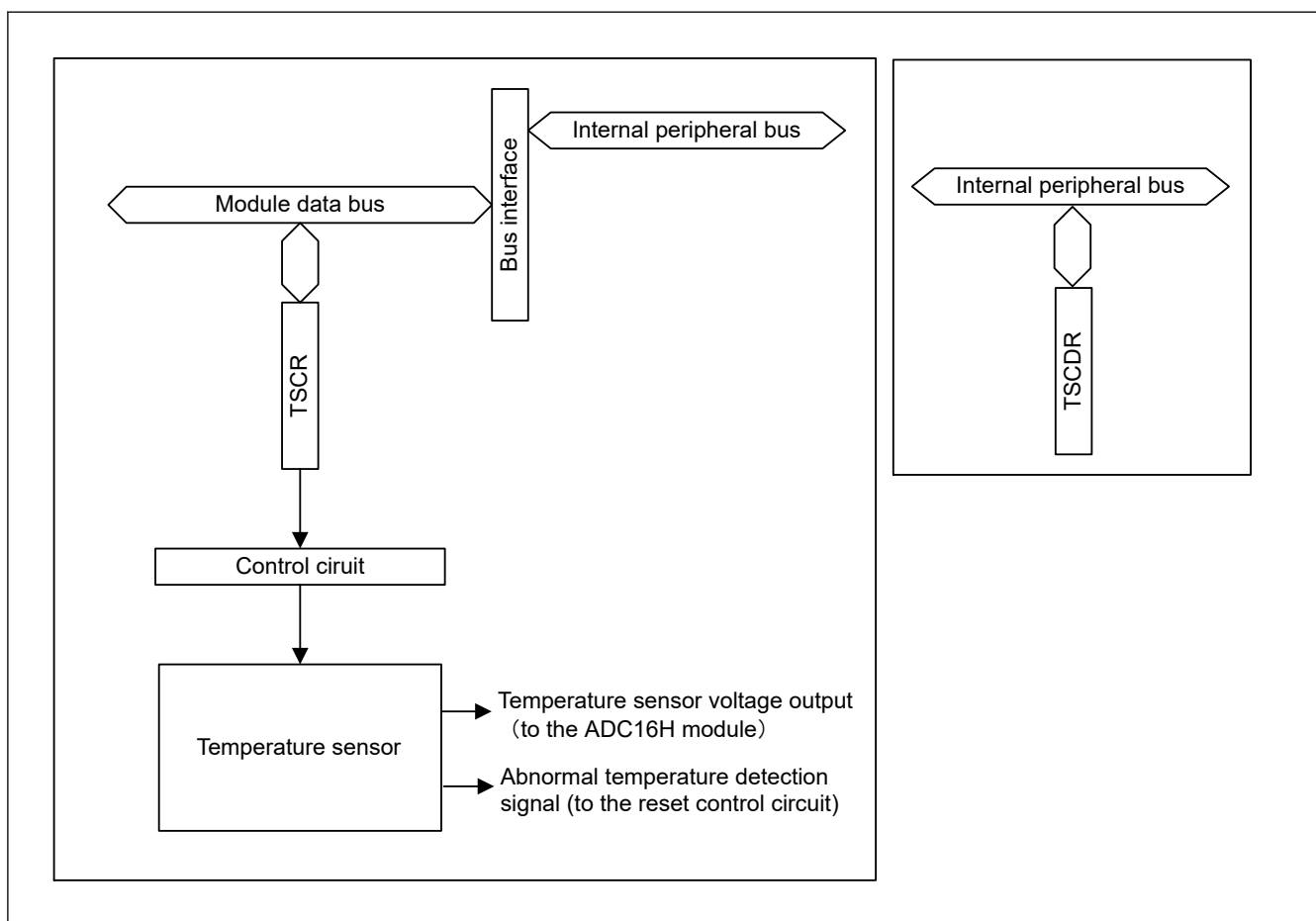


Figure 56.1 TSN block diagram

56.2 Register Descriptions

56.2.1 TSCR : Temperature Sensor Control Register

Base address: TSN = 0x4023_5000
TSN_NS = 0x5023_5000

Offset address: 0x00

Bit position:	7	6	5	4	3	2	1	0
Bit field:	TSEN	—	—	TSOE	—	—	—	—

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	TSOE	Temperature Sensor Output Enable 0: Disable output from the temperature sensor to the ADC16H 1: Enable output from the temperature sensor to the ADC16H	R/W
6:5	—	These bits are read as 0. The write value should be 0.	R/W
7	TSEN	Temperature Sensor Enable 0: Stop the temperature sensor 1: Start the temperature sensor	R/W

Note: S-TYPE3, P-TYPE3

The TSCR is a register which controls the temperature sensor. The timing constraints shown in [Figure 56.3](#) apply to the settings of the TSCR register.

TSOE bit (Temperature Sensor Output Enable)

The TSOE bit enables or disables the temperature sensor output to ADC16H.

TSEN bit (Temperature Sensor Enable)

The TSEN bit starts or stops the temperature sensor.

56.2.2 TSCDR : Temperature Sensor Calibration Data Register

Base address: TSD = 0x02C1_EDA0
TSD_NS = 0x12C1_EDA0

Offset address: 0x00

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TSCD[15:0]															

Value after reset: Chip-specific value

Bit	Symbol	Function	R/W
15:0	TSCD[15:0]	Temperature Sensor Calibration Data Chip-specific value	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE5, P-TYPE5

The TSCDR register stores temperature sensor calibration data measured for each chip at factory shipment.

Temperature sensor calibration data is the output voltage of the temperature sensor under the conditions $T_j = 125 \pm 2^\circ\text{C}$ or $105 \pm 2^\circ\text{C}$ and $AVCC0 = VREFH0 = 3.3\text{ V}$ converted to a digital value by the A/D converter. For details on the measurement temperature condition, see [section 56.3.1. Preparation for Using the Temperature Sensor](#).

The TSCDR register is a read-only 32-bit register. Read from this register in 32-bit units.

Temperature sensor calibration data is stored in the lower 12 bits of the TSCDR register.

56.2.3 TSCDR2 : Temperature Sensor Calibration Data Register 2

Base address: TSD = 0x02C1_EDA0
TSD_NS = 0x12C1_EDA0

Offset address: 0x04

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TSCD[15:0]															
Value after reset:	Chip-specific value															

Bit	Symbol	Function	R/W
15:0	TSCD[15:0]	Temperature Sensor Calibration Data Chip-specific value	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE5, P-TYPE5

The TSCDR2 register stores temperature sensor calibration data measured for each chip at factory shipment.

Temperature sensor calibration data is the output voltage of the temperature sensor under the conditions $T_j = -40 \pm 2^\circ\text{C}$ and $AVCC0 = VREFH0 = 3.3\text{ V}$ converted to a digital value by the A/D converter. For details on the measurement temperature condition, see [section 56.3.1. Preparation for Using the Temperature Sensor](#).

The TSCDR2 register is a read-only 32-bit register. Read from this register in 32-bit units.

Temperature sensor calibration data is stored in the lower 12 bits of the TSCDR2 register.

56.3 Using the Temperature Sensor

The temperature sensor outputs a voltage that varies with the temperature. This voltage is converted to a digital value by the A/D converter. To obtain the die temperature, convert this value into the temperature.

56.3.1 Preparation for Using the Temperature Sensor

The ambient temperature (T) is proportional to the temperature sensor voltage output (V_s), so ambient temperature is calculated with the following formula:

$$T = (V_s - V_1) / \text{slope} + T_1$$

- T: Ambient temperature of MCU as calculation result ($^\circ\text{C}$)
- V_s : Voltage output by the temperature sensor on temperature measurement (V)
- T_1 : Temperature experimentally measured at one point ($^\circ\text{C}$)
- V_1 : Voltage output by the temperature sensor on measurement of T_1 (V)
- T_2 : Temperature experimentally measured at a second point ($^\circ\text{C}$)
- V_2 : Voltage output by the temperature sensor on measurement of T_2 (V)
- Slope: Temperature gradient of the temperature sensor ($\text{V} / ^\circ\text{C}$), $\text{slope} = (V_2 - V_1) / (T_2 - T_1)$

Characteristics vary between sensors, so Renesas recommends measuring two different sample temperatures as follows:

1. Use the A/D converter to measure the voltage V1 output by the temperature sensor at temperature T1.
2. Again use the A/D converter to measure the voltage V2 output by the temperature sensor at a different temperature T2.
3. Obtain the temperature gradient (slope = $(V2 - V1) / (T2 - T1)$) from these results.
4. Subsequently, obtain temperatures by substituting the slope into the formula for the temperature characteristic ($T = (Vs - V1) / \text{slope} + T1$).

In order to make the temperature Ta and Tj as close as possible, perform the measurement with the power consumption of the MCU as low as possible.

If you are using the temperature gradient given in [section 70, Electrical Characteristics](#), use the A/D converter to measure the voltage V1 output by the temperature sensor at temperature T1, then calculate the temperature characteristic using the following formula:

$$T = (Vs - V1) / \text{slope} + T1$$

Note: This method produces less accurate temperatures than measurement at two points.

In this MCU, the TSCDR register and TSCDR2 register store the temperature values of the temperature sensor measured under the condition Ta = Tj = 125±2 °C or 105±2 °C, -40±2 °C and AVCC0 = VREFH0 = 3.3 V. If you use this value as the sample measurement result at the first and second points, you can omit the preparation before using the temperature sensor.

In case of Tj Max = 125 °C^{*1}, the TSCDR register and TSCDR2 register store the temperature values (CAL125 and CAL-40) of the temperature sensor measured under the condition Ta = Tj = 125±2 °C, -40±2 °C and AVCC0 = VREFH0 = 3.3 V.

V1 is calculated from CAL125 or CAL-40:

$$V1 = 3.3 \times \text{CAL125 or CAL-40} / 4096 \text{ [V]} \text{ (for 12 bits accuracy)}$$

Using this value, the measured temperature can be calculated according to the following formula:

$$T = (Vs - V1) / \text{slope} + 125 \text{ or } -40 \text{ [°C]}$$

- T: Ambient temperature of MCU as calculation result (°C)
- Vs: Voltage output by the temperature sensor when the temperature is measured (V)
- V1: Voltage output by the temperature sensor when Ta = Tj = 125 or -40°C and AVCC0 = VREFH0 = 3.3 V (V)
- Slope: Temperature gradient of the temperature sensor^{*2} / 1000 (V/°C)
Slope also can be obtained by using TSCDR register or TSCDR2 register value as the sample measurement result at the second point.

In case of Tj Max = 95 °C or 105 °C^{*1}, the TSCDR register and TSCDR2 register store the temperature values (CAL105 and CAL-40) of the temperature sensor measured under the condition Ta = Tj = 105±2 °C, -40±2 °C and AVCC0 = VREFH0 = 3.3 V. The TSCDR2 register stores the temperature value of the temperature sensor measured under the condition Ta = Tj = -40±2 °C and AVCC0 = VREFH0 = 3.3 V.

V1 is calculated from CAL105 or CAL-40.

$$V1 = 3.3 \times \text{CAL105 or CAL-40} / 4096 \text{ [V]} \text{ (for 12 bits accuracy)}$$

Using this value, the measured temperature can be calculated according to the following formula:

$$T = (Vs - V1) / \text{slope} + 105 \text{ or } -40 \text{ [°C]}$$

- T: Ambient temperature of MCU as calculation result (°C)
- Vs: Voltage output by the temperature sensor when the temperature is measured (V)
- V1: Voltage output by the temperature sensor when Ta = Tj = 105 or -40 °C and AVCC0 = VREFH0 = 3.3 V (V)
- Slope: Temperature gradient of the temperature sensor^{*2} / 1000 (V/°C)
Slope also can be obtained by using TSCDR register or TSCDR2 register value as the sample measurement result at the second point.

Note 1. The upper limit of operating junction temperature is 95 °C, 105 °C or 125 °C, depending on the product. For details, see [section 1.3, Part Numbering](#).

Note 2. See [section 70, Electrical Characteristics](#).

56.3.2 Procedures for Measuring Temperature Sensor Using the Temperature Sensor with ADC16H

[Figure 56.2](#) shows the procedure for using the TSN.

For details on the ADC16H setting procedures, see [section 54, 16-bit A/D Converter \(ADC16H\)](#). For details on the TEMPRCR register, see [section 6, Resets](#).

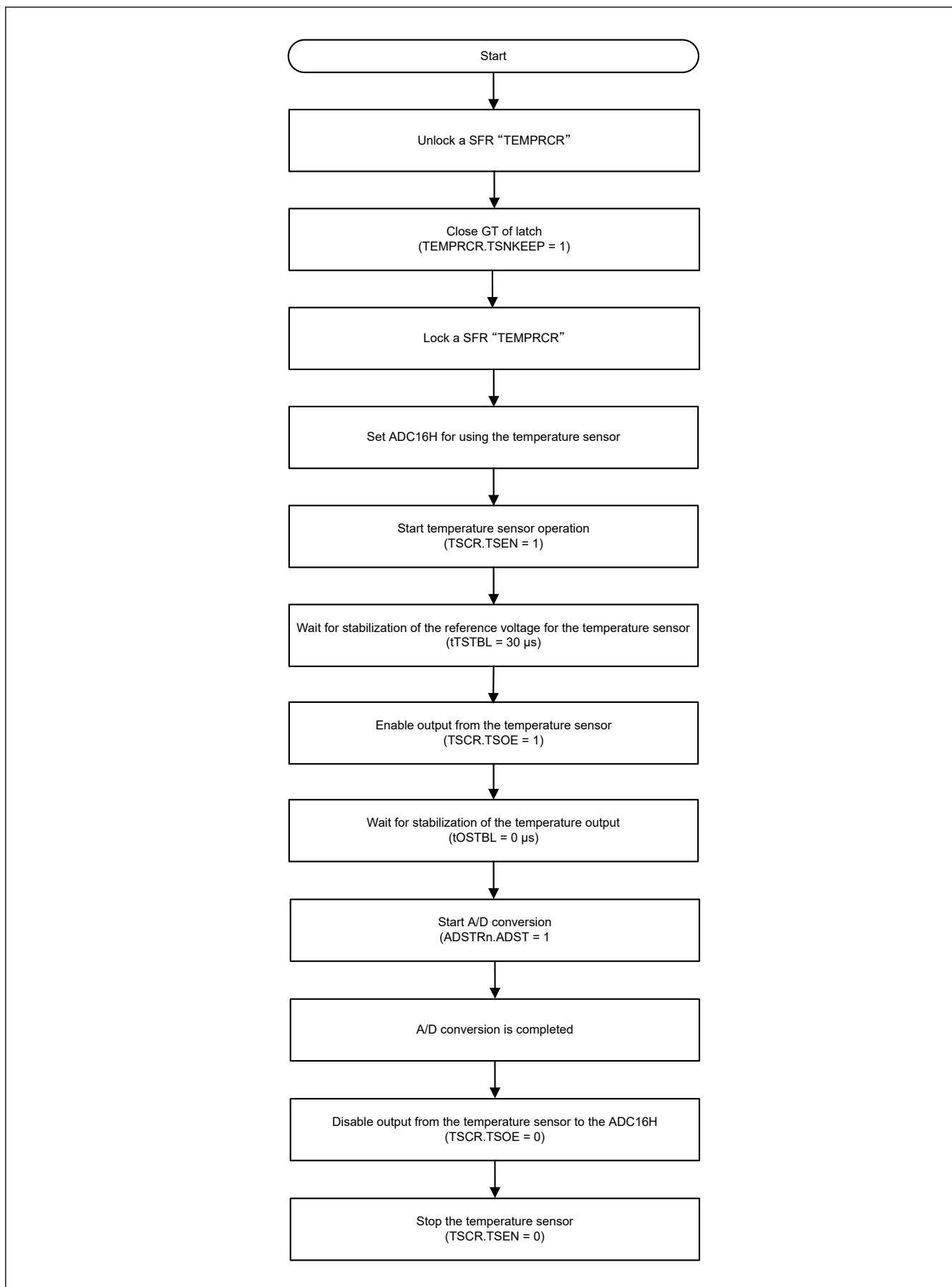
**Figure 56.2 Procedure example for measuring temperature using TSN with ADC16H**

Figure 56.3 shows the timing from the start of temperature sensor operation until the completion of A/D conversion when the ADC16H is in single scan mode (the conversion target is the temperature sensor output only). The times shown in the figure are described in Table 56.2.

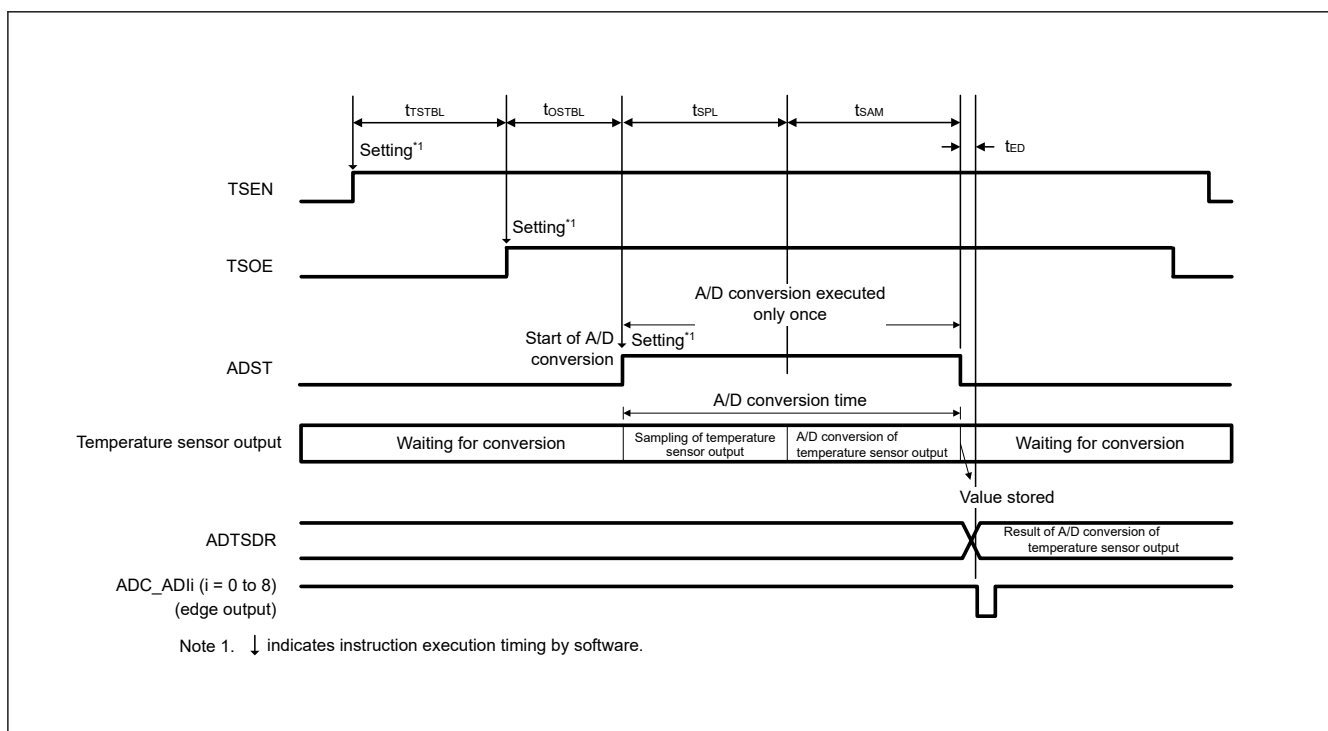


Figure 56.3 Timing from start of temperature sensor operation until completion of A/D conversion

Table 56.2 Time until completion of A/D conversion after start of temperature sensor operation

Parameter	Symbol	Time
Wait time for temperature sensor reference voltage stabilization	t_{TSTBL}	30 μ s (min)
Wait time for temperature sensor output stabilization	t_{OSTBL}	0 μ s (min)
Wait time for temperature sensor abnormal temperature detection signal stabilization	t_{RSTBL}	30 μ s (min)
A/D converter input sampling time	t_{SPL}	ADSSTRn setting $\times$ ADCLK period
A/D conversion time	t_{SAM}	See Table XX.XX, Conversion times during scanning (in numbers of cycles of ADCLK and PCLKA)
Scan conversion end delay	t_{ED}	[DEADBEEF].

56.3.3 Procedures for Using the Temperature Monitor Reset

TSN can detect abnormal temperature and generate a signal to the reset circuit. If the temperature is outside the specified detection range^{*1}, reset is issued. The function of temperature monitor reset is enabled using specific procedures that are described in Figure 56.4 and Figure 56.5.

For more information, see section 6.3.12. Temperature Monitor Reset.

Note 1. For details on the detection range of abnormal temperature, see section 70, Electrical Characteristics.

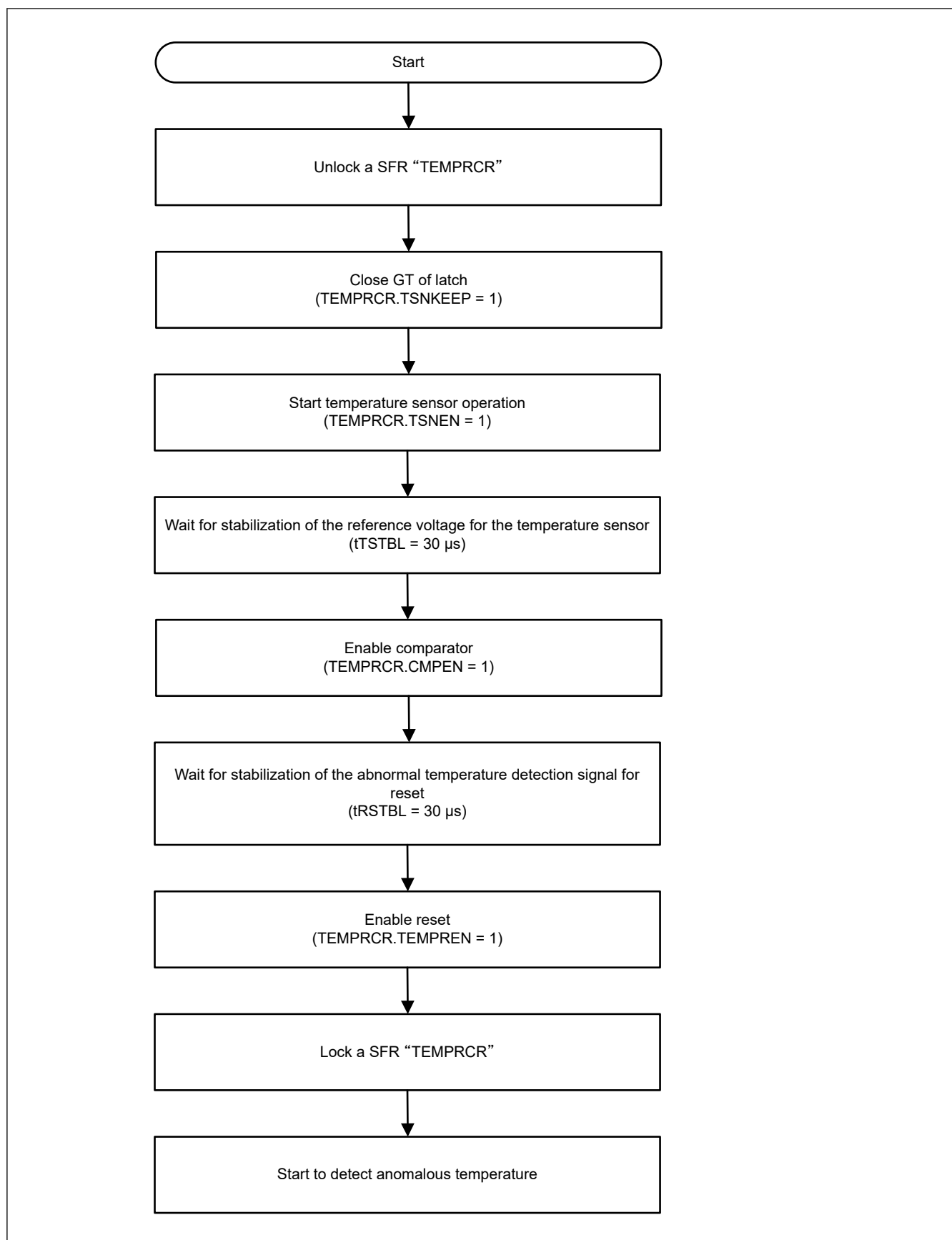


Figure 56.4 Procedure example for enabling temperature monitor reset but not measuring temperature with ADC16H

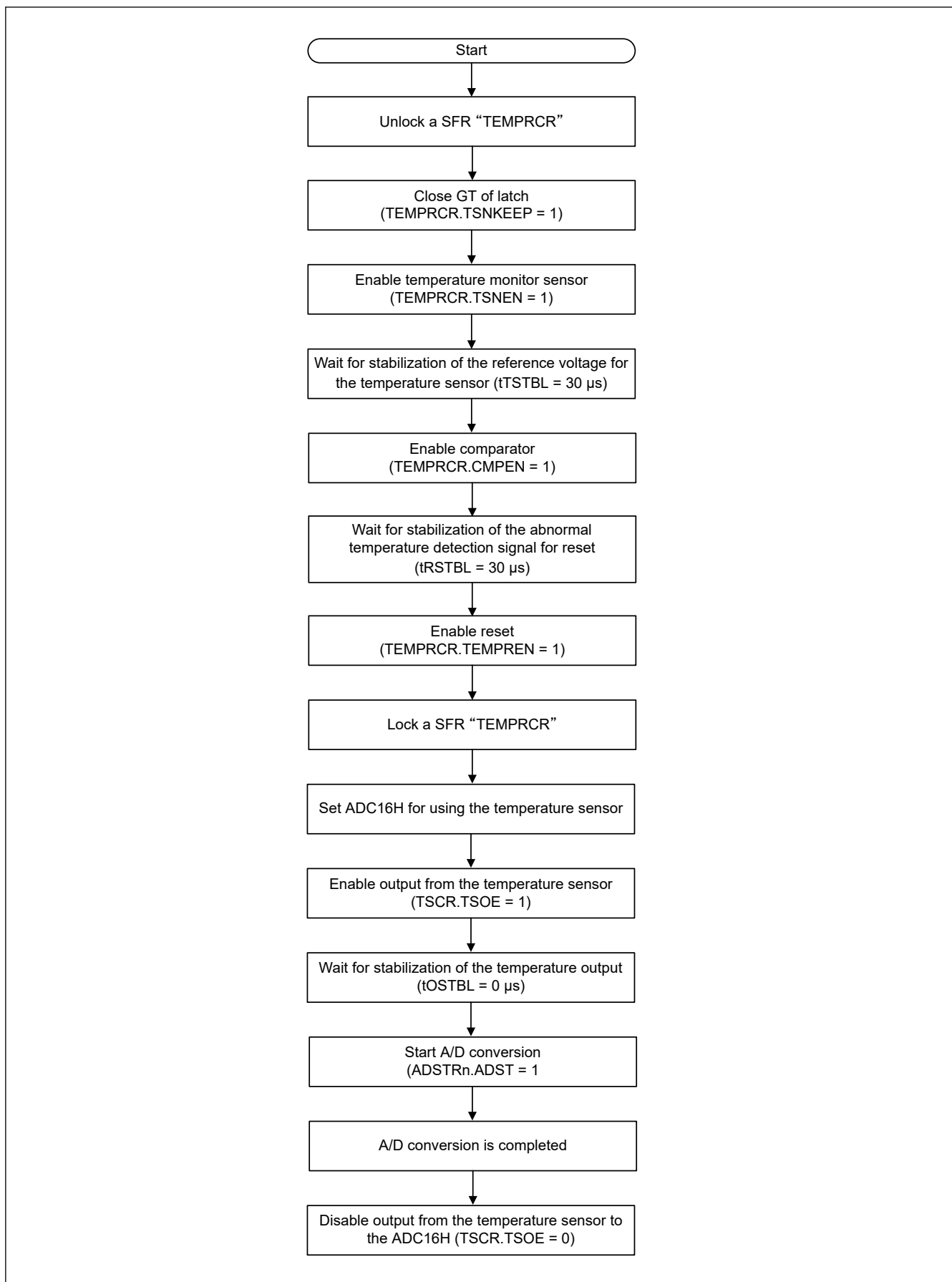


Figure 56.5 Procedure example for enabling temperature monitor reset and measuring temperature with ADC16H

56.4 Usage Notes

56.4.1 Settings for the Module-Stop Function

TSN operation can be disabled or enabled using the associated bit in Module Stop Control Register D (MSTPCRD). The TSN is initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see [section 11, Low Power Mode](#).

56.4.2 Settings for the Software Standby Mode

Before entering Software Standby mode, set TSCR.TSOE bit to 0 after stopping A/D conversion, then set TSCR.TSEN bit to 0 to stop the TSN. [Figure 56.6](#) shows the procedure for entering Software Standby mode and [Figure 56.7](#) shows the procedure for returning from Software Standby mode.

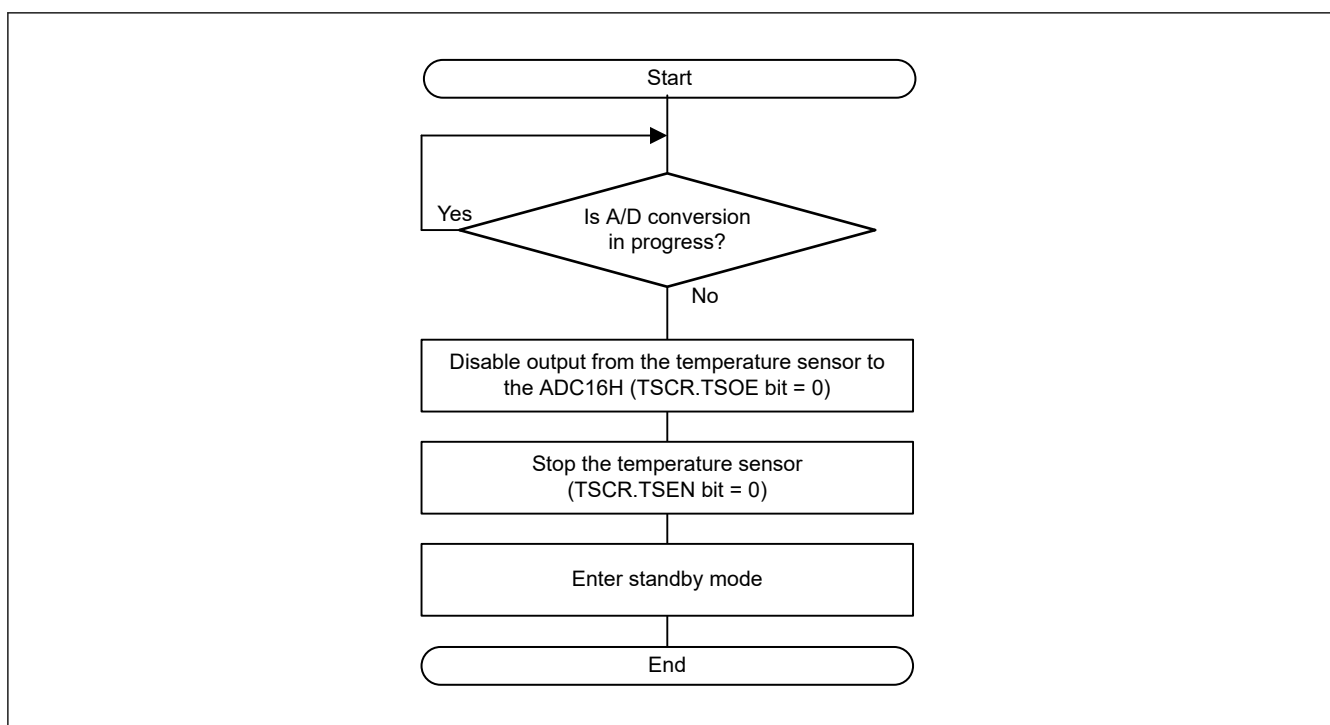


Figure 56.6 Example procedure for entering Software Standby mode

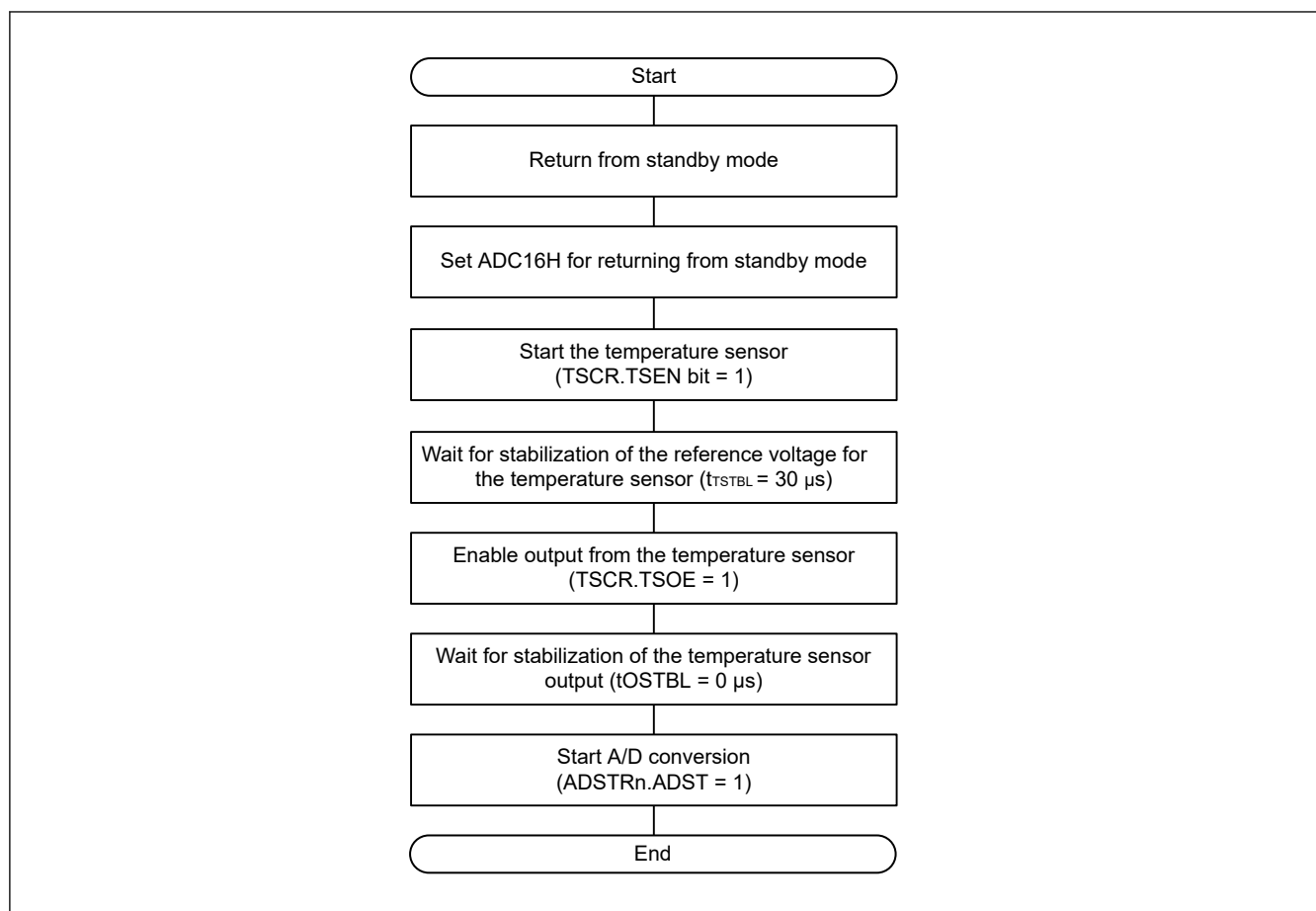


Figure 56.7 Example procedure for returning from Software Standby mode